The route is the decision.

Which fleet powertrains fit which route types, how to build the operating model, and how to plan the pilot that proves it.

Which powertrain fits which route

Fit levels are a reading of the source deck's guidance, not measured figures. "Not covered" means the deck does not address that pairing.

	Depot-based, predictable	Long-haul, high utilization	Dense urban drops	Mixed routes, uncertain charging	Existing diesel assets
Battery-electric	Strong fit	Poor fit	Conditional	Conditional	Conditional
Hybrid	Conditional	Not covered	Not covered	Strong fit	Conditional
Hydrogen fuel-cell	Not covered	Strong fit	Not covered	Not covered	Not covered
Biofuels	Not covered	Conditional	Not covered	Conditional	Strong fit
Cargo bikes	Not covered	Not covered	Strong fit	Not covered	Not covered

Reasoning behind each rated pairing

Battery-electric

- **Depot-based, predictable** (Strong fit): Charging can be planned around the operating day.
- **Long-haul, high utilization** (Poor fit): Batteries are harder to fit where distance, payload, and utilization are high.
- **Dense urban drops** (Conditional): Smaller, lighter vehicles solve a different problem at the curb.
- **Mixed routes, uncertain charging** (Conditional): Confirm charging access and route confidence first.
- **Existing diesel assets** (Conditional): Replace diesel where the route is ready.

Hybrid

- **Depot-based, predictable** (Conditional): A bridge while charging capacity is built.
- **Mixed routes, uncertain charging** (Strong fit): Useful when charging access is uncertain.
- **Existing diesel assets** (Conditional): A staged step away from diesel.

Hydrogen fuel-cell

- **Long-haul, high utilization** (Strong fit): Preserves distance, payload, and utilization.

Biofuels

- **Long-haul, high utilization** (Conditional): Liquid-fuel pathways may preserve distance and payload.
- **Mixed routes, uncertain charging** (Conditional): Keeps the system familiar while the route picture clears.
- **Existing diesel assets** (Strong fit): Lower-carbon bridge that reduces disruption.

Cargo bikes

- **Dense urban drops** (Strong fit): Cargo bikes own the last 500 meters.

Powertrain profiles

Battery-electric

Best fit. Return-to-depot routes: regional delivery and predictable daily mileage.

Why it works. High drivetrain efficiency, quiet operation, zero tailpipe emissions.

Watch-outs. Plan for charging capacity, dwell time, payload trade-offs, and winter performance.

Hybrid

Best fit. Mixed routes, uncertain charging access, or a fleet starting a staged transition.

Why it works. Shift more work to electric operation as charging access and route confidence improve.

Watch-outs. Treat it as a bridge with a clear replacement trigger, not the final state.

Hydrogen fuel-cell

Best fit. Long distances, high payloads, and fast refueling requirements.

Why it works. Preserves operating patterns that are harder to electrify with batteries.

Watch-outs. Economics depend on reliable access to low-carbon hydrogen and fueling infrastructure.

Biofuels

Best fit. Existing diesel assets where replacement is not yet practical.

Why it works. Vehicles, fueling systems, and operating practices stay familiar to the team.

Watch-outs. Check feedstock availability, lifecycle emissions, supply certainty, and long-term lock-in.

Cargo bikes

Best fit. Dense urban zones, short drops, congestion, and curb-constrained streets.

Why it works. Shift parcels to a microhub, then complete deliveries with cargo bikes.

Watch-outs. Plan for payload, weather, battery swaps, rider safety, and secure handoff points.

Operating model

Powertrain selection is the start of a system decision.

1	2	3	4
Segment routes	**Assign powertrain**	**Build capacity**	**Measure performance**
By distance, payload, dwell time, and access constraints.	Give each segment the lowest-risk viable powertrain.	Charging, fueling, microhub, and maintenance.	Cost, service, utilization, energy, and emissions.

Pilot tracker

Work through each stage in order and tick items off as they are completed.

Baseline 7 items

Capture how the work runs today.

- [ ] Routes
- [ ] Mileage
- [ ] Payload
- [ ] Stops
- [ ] Dwell
- [ ] Fuel
- [ ] Service levels

Pilot 3 items

Cover the range of the work.

- [ ] A representative depot
- [ ] A demanding regional route
- [ ] One urban zone

Scale 3 items

Expand only when all three clear the bar.

- [ ] Total cost
- [ ] Uptime
- [ ] Customer experience